

BREADTH FIRST SEARCH

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

int a[20][20],q[20],visited[20],n,i,j,f=0,r=-1;

void bfs(int v)

{
    for(i=1;i<=n;i++)
        if(a[v][i] && !visited[i])
            q[++r]=i;
        if(f<=r)
        {
            visited[q[f]]=1;
            bfs(q[f++]);
        }
    }

void main()

{
    int v;

    clrscr();

    printf("\n Enter the number of vertices:");

    scanf("%d",&n);

    for(i=1;i<=n;i++)
    {
        q[i]=0;
```

```
visited[i]=0;

}

printf("\n Enter graph data in matrix form:\n");

for(i=1;i<=n;i++)

for(j=1;j<=n;j++)

scanf("%d",&a[i][j]);

printf("\n Enter the starting vertex:");

scanf("%d",&v);

bfs(v);

printf("\n The node which are reachable are:\n");

for(i=1;i<=n;i++)

if(visited[i])

printf("%d\t",i);

getch();

}
```